Practice_Problem_01

Nafis Mohammad

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```
# Load the required packages
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at C:/Users/nafis/OneDrive/Documents/Data_Mining_Modeling_Biostat/Practice_Problem_01
```

```
# Import the dataset
heart <- read.csv(
  "C:/Users/nafis/OneDrive/Documents/Data_Mining_Modeling_Biostat/Practice_Problem_01/healthcare-dataset
  header = TRUE,
  stringsAsFactors = TRUE
)

# Examine the two variables
heart %>%
  select(bmi, avg_glucose_level) %>%
  head()
```

```
##    bmi avg_glucose_level
## 1 36.6            228.69
## 2  N/A            202.21
## 3 32.5            105.92
## 4 34.4            171.23
## 5   24            174.12
## 6   29            186.21
```

> "BMI is the independent variable because it is being used to predict average glucose level, which is the dependent variable. BMI is easier to clinically evaluate because it only requires a patient's height and weight, whereas measuring blood glucose requires a blood sample or specialized glucose-monitoring equipment."

```r
# Check the structure and missing values
str(heart)
```

```
## 'data.frame':    5110 obs. of  12 variables:
##  $ id               : int  9046 51676 31112 60182 1665 56669 53882 10434 27419 60491 ...
##  $ gender           : Factor w/ 3 levels "Female","Male",..: 2 1 2 1 1 2 2 1 1 1 ...
##  $ age              : num  67 61 80 49 79 81 74 69 59 78 ...
##  $ hypertension     : int  0 0 0 0 1 0 1 0 0 0 ...
##  $ heart_disease    : int  1 0 1 0 0 0 1 0 0 0 ...
##  $ ever_married     : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2 2 1 2 2 ...
##  $ work_type        : Factor w/ 5 levels "children","Govt_job",..: 4 5 4 4 5 4 4 4 4 4 ...
##  $ Residence_type   : Factor w/ 2 levels "Rural","Urban": 2 1 1 2 1 2 1 2 1 2 ...
##  $ avg_glucose_level: num  229 202 106 171 174 ...
##  $ bmi              : Factor w/ 419 levels "10.3","11.3",..: 240 419 199 218 114 164 148 102 419 116
##  $ smoking_status   : Factor w/ 4 levels "formerly smoked",..: 1 2 2 3 2 1 2 2 4 4 ...
##  $ stroke           : int  1 1 1 1 1 1 1 1 1 1 ...
```

```r
summary(heart)
```

```
##        id            gender          age         hypertension
##  Min.   :   67   Female:2994   Min.   : 0.08   Min.   :0.00000
##  1st Qu.:17741   Male  :2115   1st Qu.:25.00   1st Qu.:0.00000
##  Median :36932   Other :   1   Median :45.00   Median :0.00000
##  Mean   :36518                 Mean   :43.23   Mean   :0.09746
##  3rd Qu.:54682                 3rd Qu.:61.00   3rd Qu.:0.00000
##  Max.   :72940                 Max.   :82.00   Max.   :1.00000
##
##  heart_disease     ever_married         work_type    Residence_type
##  Min.   :0.00000   No :1757     children     : 687   Rural:2514
##  1st Qu.:0.00000   Yes:3353     Govt_job     : 657   Urban:2596
##  Median :0.00000                Never_worked :  22
##  Mean   :0.05401                Private      :2925
##  3rd Qu.:0.00000                Self-employed: 819
##  Max.   :1.00000
##
##  avg_glucose_level      bmi               smoking_status     stroke
##  Min.   : 55.12    N/A    : 201   formerly smoked: 885   Min.   :0.00000
##  1st Qu.: 77.25    28.7   :  41   never smoked   :1892   1st Qu.:0.00000
##  Median : 91.89    28.4   :  38   smokes         : 789   Median :0.00000
##  Mean   :106.15    26.1   :  37   Unknown        :1544   Mean   :0.04873
##  3rd Qu.:114.09    26.7   :  37                          3rd Qu.:0.00000
##  Max.   :271.74    27.6   :  37                          Max.   :1.00000
##                    (Other):4719
```

```r
# Data wrangling
heart_clean <- heart %>%
  select(bmi, avg_glucose_level) %>%
  mutate(bmi = as.numeric(as.character(bmi))) %>%
  drop_na(bmi, avg_glucose_level)
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `bmi = as.numeric(as.character(bmi))`.
```

```
## Caused by warning:
## ! NAs introduced by coercion
```

```
# Check the cleaned dataset
str(heart_clean)
```

```
## 'data.frame':    4909 obs. of  2 variables:
##  $ bmi              : num  36.6 32.5 34.4 24 29 27.4 22.8 24.2 29.7 36.8 ...
##  $ avg_glucose_level: num  229 106 171 174 186 ...
```

```
nrow(heart_clean)
```

```
## [1] 4909
```

```
# Run a linear regression model
glucose_model <- lm(avg_glucose_level ~ bmi, data = heart_clean)

# View the results
summary(glucose_model)
```

```
##
## Call:
## lm(formula = avg_glucose_level ~ bmi, data = heart_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -111.84  -28.62  -12.43   10.95  164.65
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 76.62344    2.38012   32.19   <2e-16 ***
## bmi          0.99268    0.07949   12.49   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 43.74 on 4907 degrees of freedom
## Multiple R-squared:  0.0308, Adjusted R-squared:  0.0306
## F-statistic: 155.9 on 1 and 4907 DF,  p-value: < 2.2e-16
```

> "Before running the linear regression model, the BMI and average glucose level columns were selected. BMI was converted into a numeric variable, and observations containing missing values were removed, leaving 4,909 observations. A linear regression model was performed using BMI as the independent variable and average glucose level as the dependent variable. The results showed a statistically significant positive relationship between BMI and average glucose level ($p < 0.05$). However, the R-squared value of 0.0308 indicates that BMI explains only approximately 3.08% of the variation in average glucose levels. Therefore, although BMI is a statistically significant predictor of average glucose level, the relationship is relatively weak."

```
# Calculate the slope
slope <- sum(
  (heart_clean$bmi - mean(heart_clean$bmi)) *
```

```
  (heart_clean$avg_glucose_level - mean(heart_clean$avg_glucose_level))
) / sum(
  (heart_clean$bmi - mean(heart_clean$bmi))^2
)

# Calculate the intercept
intercept <- mean(heart_clean$avg_glucose_level) -
  slope * mean(heart_clean$bmi)

# Create a function to predict glucose level from BMI
predict_glucose <- function(bmi) {
  predicted_glucose <- intercept + slope * bmi
  return(predicted_glucose)
}

# Test the function using a BMI of 25
predict_glucose(25)
```

```
## [1] 101.4404
```

```
# Test the function using a BMI of 30
predict_glucose(30)
```

```
## [1] 106.4038
```

> "A function called predict_glucose was created using the slope and intercept of the linear regression equation to predict average glucose level from BMI. The function estimates an average glucose level of 101.44 for a patient with a BMI of 25 and 106.40 for a patient with a BMI of 30. The positive slope indicates that predicted average glucose level increases by approximately 0.99 units for every one-unit increase in BMI. However, because the correlation is weak, BMI alone is not a strong predictor of glucose level."